

Assignment - 03

22BCN1068

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Subject: Computer Graphics

Section - A

1. What is the use of homogeneous coordinates and matrix representation?

Ans Homogeneous coordinates and matrix representation are used primarily in computer graphics, computer vision, and geometric transformations to simplify and unify the representation of transformations, especially in higher-dimensional spaces like 2D and 3D.

2. What two-dimensional rotation equation in the matrix form?

Ans The 2D rotation matrix for an angle θ is :-

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

3. What is transformation?

Ans In computer graphics, transformation refers to the process of changing the position, size, orientation or shape of objects in a scene. This can include operations like translation (moving), scaling (resizing), rotation and shearing.

4. Define scaling in transformation?

Ans Scaling in transformation refers to changing the size of an object by increasing or decreasing its dimensions, either uniformly (same factor in all directions) or non-uniformly (different factor in each direction).

5. Explain Reflection in transformation.

Ans Reflection in transformation is a type of geometric transformation that flips a shape over a line, creating a mirror image of the original shape. Each point of the

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Shape are mapped to their corresponding points on the opposite side of the line.

Reflection across X-axis - $\begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix}$

Reflection across Y-axis - $\begin{vmatrix} -1 & 0 \\ 0 & 1 \end{vmatrix}$

• Section - B

Q. What do you mean by transformation?

Ans. In computer graphics, a transformation is a mathematical function that maps one set of points (the input) to another set of points (the output). This mapping is used to manipulate objects in 2D or 3D space, allowing us to create complex scenes and animations. Common types of transformations include:

- Translation :- Shifting an object to a new location.
- Rotation :- Rotating an object around a specific point.
- Scaling :- Enlarging or shrinking an object.
- Shearing :- Skewing an object along a particular axis.
- Projection :- Projecting a 3D object onto a 2D screen.

Above all five methods are part of transformation and all are coming in transformation function.

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2. Why do you need transformations in computer graphics?
 Ans: Transformations are essential in computer graphics for several reasons:

Creating complex scenes:- By combining multiple transformations we can create intricate and realistic scene with objects of various shapes and sizes.

Animating objects:- By applying transformations over time, we can animate objects, making them move, rotate and change shape.

- Simulating real-world phenomena:- Transformation can be used to simulate physical phenomena like gravity, collisions, and deformations.
- Interacting with virtual objects:- Transformations allow users to manipulate virtual objects such as rotating a 3D model or zooming in on a 2D image.
- Rendering images:- Transformations are used to project 3D objects onto a 2D screen, creating the illusion of depth and perspective.

3. What is the need of homogeneous coordinates?

Ans: Homogeneous coordinates are a mathematical technique that extends the traditional Cartesian coordinate system by adding an extra dimension. This extra dimension allows us to represent various geometric transformations, such as translation, rotation, scaling and shearing, using a single matrix operation.

The key advantages of using homogeneous coordinates are:-

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- Unified Representation of transformation :- All transformations can be represented as matrix multiplications, simplifying complex sequences of transformations.
- Efficient Computations :- Matrix operations are highly optimized and can be efficiently implemented in hardware, making them suitable for real-time graphics applications.
- Projection and Perspective :- Homogeneous coordinates are essential for projecting 3D objects onto a 2D screen, creating realistic perspective effects.
- Geometric Calculations :- They facilitate various geometric calculations, such as finding intersections, distances, and angles between lines and curves, in a concise and elegant manner.

By using homogeneous coordinates we can simplify the implementation of complex graphics algorithms and achieve more efficient and accurate results.

• Sec - c ($3 \times 11 = 33$)

1. What are the basic transformations? Describe each with their matrix representation.

Ans Basic transformations in computer graphics involve manipulating objects in a 2D or 3D coordinate system. The primary transformations are translation, scaling, rotation and shearing. Each transformation can be represented using a matrix, which allows for combining multiple transformations into a single matrix for efficient computation.

1. Translation :- Translation involves shifting an object by a certain distance along the x, y and z (in 3D) axes.

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2D Translation Matrix: -

$$\begin{bmatrix} 1 & 0 & T_x \\ 0 & 1 & T_y \\ 0 & 0 & 1 \end{bmatrix}$$

- T_x and T_y are the distances to move the object along the x and y axes, respectively.
- A point (x, y) after translation becomes:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & T_x \\ 0 & 1 & T_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + T_x \\ y + T_y \\ 1 \end{bmatrix}$$

2. Scaling: Scaling changes the size of an object by stretching or shrinking it along the x, y (and z) axes.

2D Scaling Matrix: -

$$\begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- S_x and S_y are the scaling factors along the x and y axes.
- A point (x, y) after scaling becomes:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} S_x \cdot x \\ S_y \cdot y \\ 1 \end{bmatrix}$$

2. Rotation: Rotation involves rotating an object around a specified axis by a certain angle.

2D Rotation Matrix (angle θ): -

$$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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- A point (x, y) after rotation becomes:-

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \\ 1 \end{bmatrix}$$

- Shearing:- shearing skews the shape of an object, effectively distorting it by shifting one axis while keeping the other fixed.

2D shearing Matrix:-

$$\begin{bmatrix} 1 & S_x & 0 \\ S_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- 2. What do you mean by reflection and shearing? Explain with examples.

Ans: Reflection is a transformation that flips an object over a specified line (axis), creating a mirror image of it.

Examples:

- (i) Reflection across the X-axis:-

• Matrix -

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

If a point (x, y) is reflected across the X-axis, it becomes $(x, -y)$.

- (ii) Reflection across the Y-axis:-

• Matrix -

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

• A point (x, y) becomes $(-x, y)$.

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(iii) Reflection across the line $y = x$.

Matrix:
$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

• A point (x, y) transforms to (y, x) .

• Use Case:- Reflection is often used in graphics to create symmetrical designs or to generate mirrored versions of objects.

• Shearing:- Shearing is a transformation that distorts the shape of an object by shifting its coordinates along one axis, while keeping the other axis fixed. It results in a slanted or skewed version of the original shape.

Examples:

(i) Horizontal Shearing (shifts along the x-axis):-

Matrix:-
$$\begin{bmatrix} 1 & S_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

• A point (x, y) becomes $(x + S_x \cdot y, y)$.

• For $S_x = 1$, a square turns into a parallelogram slanted to the right.

(ii) Vertical Shearing (shifts along the y-axis):

Matrix:-
$$\begin{bmatrix} 1 & 0 & 0 \\ S_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

• A point (x, y) becomes $(x, y + S_y \cdot x)$.

• For $S_y = 1$, a square turns into a parallelogram slanted upwards.

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Use Case:- Shearing is useful for creating effects like italics in text, or slanting shapes for perspective views in graphics.

3. Given a triangle with points $(1,1)$, $(0,0)$ and $(1,0)$. Apply shear of 2 units on x axis and 2 units on y axis. What are the new coordinates of the object?

b. Given a triangle with corner coordinates $(0,0)$, $(3,0)$ and $(3,3)$. Rotate the triangle by 90-degree in anticlockwise dirⁿ. Find out the new coordinates of the triangle after translating it by 2 units in both x and y directions.

Ans (a) Applying Shearing Transformation.
Given vertices.

A $(1,1)$

B $(0,0)$

C $(1,0)$

Apply shearing with factors of 2 unit on both the x and y axis.
matrices:-

(i) Horizontal shear (x-axis):-

$$\begin{bmatrix} 1 & S_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{where } S_x = 2$$

(ii) Vertical shear (y-axis):-

$$\begin{bmatrix} 1 & 0 & 0 \\ S_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{where } S_y = 2$$

After combined:-

matrix:-

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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Now

Using the combined matrix we can find the new coordinates by multiplying the matrix with each Vector:-

$$\begin{bmatrix} x' \\ y' \\ z \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + 2y \\ 2x + y \\ 1 \end{bmatrix}$$

(i) For Point A(1,1):-

$$x' = 1 + 2(1) = 3$$

$$y' = 2(1) + 1 = 3$$

So $A'(3,3)$

(ii) For Point B(0,0)

$$x' = 0 + 2(0) = 0$$

$$y' = 2(0) + 0 = 0 \quad \text{So } B'(0,0)$$

(iii) For point C(1,0):-

$$x' = 1 + 2(0) = 1$$

$$y' = 2(1) + 0 = 2 \quad \text{So } C'(1,2)$$

So coordinates are $A'(3,3)$, $B'(0,0)$, & $C'(1,2)$.

(b) Rotation and Translation:

Given vertices

$$A(0,0)$$

$$B(3,0)$$

$$C(3,3)$$

Step-1: The rotation matrix for a 90° counterclockwise rotation is

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

New coordinates:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$$

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(i) for point $A(0,0)$:-

$$x' = 0$$

$$y' = 0$$

So $A'(0,0)$

(ii) for point $B(3,0)$:-

$$x' = -0 = 0$$

$$y' = 3 \quad \text{So } B'(0,3)$$

(iii) for point $C(3,3)$:-

$$x' = -3$$

$$y' = 3 \quad \text{So } C'(-3,3)$$

New coordinates after Rotation:-

$$A'(0,0), B'(0,3), \text{ \& } C'(-3,3)$$

Step-2: Translation by 2 units in both x and y dirⁿ.

The matrix is

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

To translate each point by $T_x = 2, T_y = 2$:

$$(x', y') \rightarrow (x' + 2, y' + 2)$$

(i) for point $A'(0,0)$:-

$$x'' = 0 + 2 = 2$$

$$y'' = 0 + 2 = 2 \quad \text{So } A''(2,2)$$

(ii) for point $B'(0,3)$:-

$$x'' = 0 + 2 = 2$$

$$y'' = 3 + 2 = 5 \quad \text{So } B''(2,5)$$

(iii) for point $C'(-3,3)$:-

$$x'' = -3 + 2 = -1$$

$$y'' = 3 + 2 = 5 \quad \text{So } C''(-1,5)$$

So coordinates after Translation:-

$$A''(2,2)$$

$$B''(2,5)$$

$$C''(-1,5)$$

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